Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Manual Assembly Quality Assistant and Knowledge Engine			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Testing / Inspection / Analysis Techniques	Human Resource Competency Development Services	Human–System Interaction Technologies
Concept of Proposed Technology	The concept leverages computer vision and modern AI algorithms to enable advanced assistive systems for manual assembly tasks. Scene analysis tracks workers' movements through pose estimation and activity recognition, while object detection captures the status of assembled components, enabling real-time identification of process deviations and providing immediate feedback to the worker to ensure quality and reduce rework. But the system not only covers real-time quality control, it also extracts and preserves expert knowledge through video analysis and pose estimation and serves as a learning platform for new employees. Providing both deviation feedback and expert knowledge improves workers' skills on the job and helps them quickly become familiar with new processes. The system can be designed with a modular architecture comprising quality-check, knowledge base, and training modules, allowing phased deployment and easy customization per SME.			

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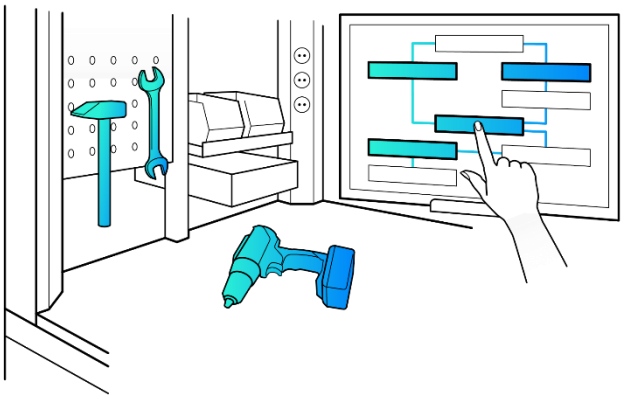
	Using standard RGB cameras minimizes hardware costs, while edge-based processing ensures low latency and reduces dependency on cloud infrastructure. This makes the solution suitable for different SMEs with strict security, data privacy and infrastructure constraints. Privacy-preserving data handling through anonymization also serves this purpose.  Fig 1: Exemplary sketch of a workplace with an advanced worker assistance system.
Rationale for Joint R&D Project	The joint R&D project addresses growing demand for AI-assisted manufacturing solutions in SMEs to close productivity and skills gaps, aligning with our core concept of an intelligent, vision-based manual assembly assistant that uses pose estimation, activity recognition, and object detection to monitor processes and provide feedback in real time. The knowledge module, which automatically extracts and preserves expert knowledge, serves as a response to demographic change and skills shortages. The rapid transformation of industry underscore the need for scalable, low-cost automation that can be rapidly deployed across diverse industries, a requirement our edge-processing, RGB-camera approach directly supports by minimizing hardware costs and latency while protecting data privacy. Public funding and policy priorities increasingly favor collaborative R&D to accelerate technology diffusion; our modular architecture (with quality-check, knowledge base, and training modules) facilitates phased deployment and shared development, allowing standardization of interfaces, data formats, and KPIs to enable wider SME adoption. Shared access to facilities, pilot lines, and real-world testbeds enhances validation, reduces individual risk, and accelerates learning curves, while multi-stakeholder collaboration improves interoperability and governance across AI, computer vision, robotics, and MES/ERP integrations. By pooling expertise and resources, the project can translate research into commercially viable, scalable solutions that bolster productivity, skill development, and national competitiveness.
Key Words	1. AI- and vision based Scene Analysis 2. Quality Assurance 3. Knowledge Preservance 4. Process documentation 5. Learning on the job

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 5	End of the Project	Level 7

R&D Objectives

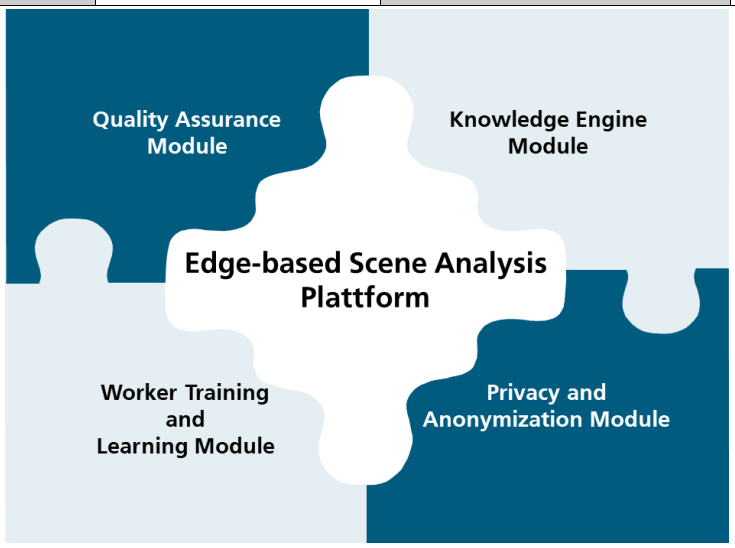

Fig 2: Overview of the different modules of the advanced worker assistance system.

Objective 1: Requirements, ethics, and compliance

- Requirements analysis with stakeholders and conduction of user needs assessments.
- A comprehensive data privacy and security policy tailored to edge and on-premises processing.
- Role-based access control, data anonymization techniques, and data retention schedules implemented in the platform.
- An ethics and compliance framework aligned with target markets' regulations and industry-specific standards.

Expected results: Demonstrated compliance through a completed security audit and risk assessment; Reduced data exposure risk with verifiable access controls and anonymization pipelines; Clear governance documents ready for external reviews and funding reporting.

Objective 2: Edge-based Scene Analysis Platform

- A production-grade edge processing platform for RGB-camera streams implementing Scene Analysis, Pose Estimation, and Activity Recognition in real time based on existing work at Fraunhofer IPA.
- Edge-integrated Object Detection to determine the status of assembled components on the edge.

Expected results: Target latency defined (ms per frame) with verified performance on standard hardware; Accurate real-time tracking of worker movements, task steps and status of assembled components with acceptable precision/recall metrics; A deployable edge stack with clear hardware requirements and setup guidelines.

Objective 3: Modules for Quality Assurance and Knowledge Engine

- An adapted Quality-Check module capable of automated deviation detection against defined standard processes.
- A Knowledge Base with structured data, a searchable repository of best practices, and a training module that surfaces curated guidance.

Expected results: Quantified improvement in deviation detection accuracy and faster issue identification; A usable knowledge portal with training material linked to real tasks, enabling rapid upskilling; Documentation of SOPs and best practices derived from pilot data.

Objective 4: Modules for Worker Training and Learning System

	• Scalable, on-the-job feedback scenarios and formative training workflows integrated into the platform. • An anonymized analytics module to extract learnings from observed practices for continuous improvement. • Adapted User Interface based on current iteration at Fraunhofer IPA. Expected results: Measurable improvement in operator proficiency (training time, error rates) during pilots; A library of anonymized best-practice patterns enabling cross-operator and cross-site learning; Evidence-based recommendations for process optimization cycles. **Objective 5: Interoperability and Integrations** • Interoperable interfaces with MES/ERP systems of SME partners. • Standardized data formats, API specifications, and KPI dashboards for broad industry adoption. Expected results: Seamless data exchange with at least two target MES/ERP systems demonstrated in pilots; A documented API surface and data schema enabling faster integration by SMEs; KPI dashboards (quality, throughput, training progress, knowledge usage) ready for deployment. **Objective 6: Pilot deployment and validation** • Field tests conducted in SME partner plants. • End-to-end validation of the platform under real production conditions. Expected results: Quantified improvements in quality metrics (e.g. rework percentage), throughput, and training outcomes; Case studies and quantified ROI estimates per site; A robust validation report with lessons learned and recommendations for scaling.

R&D Target Specifications	• Deviation detection
	  ○ Deviation detection rate: ≥ 95% in defined target processes.
	• Locations
	  ○ 2–3 SME partner facilities across diverse industries; at least two different assembly layouts.
	• Success criteria
	  ○ Quality improvement: ≥ 5% FPY (First Pass Yield).
	  ○ Training time reduction: ≥ 25%.
	  ○ Rework reduction: ≥ 50% reduction in rework rate.
	  ○ Workplace satisfaction improvement by 2 points on Likert Scale

R&D Plan	**Work package 1: Requirements, ethics, and compliance** • Task 1.1: Conduction of user needs assessments and identification of process requirements. • Task 1.2: Requirements elicitation for data privacy, security, and compliance in target markets (Korea-based SMEs; electronics/CE sectors). • Task 1.3: Design and implement a comprehensive data privacy and security policy for edge/on-prem processing. • Task 1.4: Design role-based access control, data anonymization, and data retention mechanisms. • Task 1.5: Develop an ethics and compliance framework aligned with local regulations and industry standards. • Task 1.6: Conduct security risk assessment and prepare documentation for external audits. Methodologies: privacy-by-design, risk assessment framework, data driven user needs assessments using structured interviews and eye tracking, privacy impact assessments, GDPR-like and local regulatory mapping. Milestones: Milestone 1.1: Requirements and policy draft completed (Month 2). Milestone 1.2: Core security controls implemented (Month 4). Milestone 1.3: Ethics/compliance framework approved (Month 5). Milestone 1.4: Security audit readiness and risk assessment completed (Month 6).

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Work package 2: Edge-based Scene Analysis Platform

- Task 2.1: Refine edge-processing stack for real-time Scene Analysis, Pose Estimation, and Activity Recognition based on Fraunhofer IPA work.
- Task 2.2: Integrate edge-embedded Object Detection for component status and deviation detection.
- Task 2.3: Optimize latency, accuracy, and resource usage on representative hardware.
- Task 2.4: Develop deployment guidelines, hardware requirements, and setup procedures.

Methodologies: state of the art image-based Activity recognition, hand pose estimation and object detection models, edge AI optimization (quantization, pruning), real-time computer vision, benchmark on representative hardware, continuous integration for edge builds.

Milestones:

Milestone 2.1: Baseline edge platform with core modules validated on hardware (Month 6).

Milestone 2.3: Latency/accuracy targets met on target hardware (Month 12).

Milestone 2.4: Deployment guide and hardware bill of materials published (Month 12).

Work package 3: Modules for Quality Assurance (QA) and Knowledge Engine

- Task 3.1: Data collection and annotation for project specific use cases.
- Task 3.2: Fine-tuning of state of the art models.
- Task 3.3: Adapt and implement Quality-Check module for automated deviation detection.
- Task 3.4: Develop Knowledge Base structure, taxonomy, and a training module with curated guidance.
- Task 3.5: Extract and codify best practices, including identification, consolidation, validation, and documentation of transferable practices for knowledge reuse.
- Task 3.6: Link QA outputs to knowledge content (capturing learnings as best practices).

Methodologies: fine-tuning of ai models, model-based deviation detection, pose deviation detection.

Milestones:

Milestone 3.1: QA module prototype demonstrated (Month 12).

Milestone 3.2: Knowledge Base prototype with search and training module (Month 18).

Milestone 3.3: Integration of pilot system and first SOPs/best practices (Month 24).

Work package 4: Modules for Worker Training and Learning System

- Task 4.1: Design scalable, on-the-job feedback scenarios and formative training workflows.
- Task 4.2: Develop anonymized analytics to extract learnings for continuous improvement.
- Task 4.3: Implement User Interface, using human-centered design.
- Task 4.4: Validate training impact (proficiency gains, training time reductions).

Methodologies: step-by-step guidance for on-the-job training and breakdown of processes in modules, data anonymization pipelines, user needs assessments for human-centered design.

Milestones:

Milestone 4.1: Training scenarios and workflows defined (Month 12).

Milestone 4.2: Anonymized analytics module prototype (Month 18).

Milestone 4.3: UI/UX iteration cycle completed (Month 24).

Work package 5: Interoperability and Integrations

- Task 5.1: Define and implement interfaces to MES/ERP of pilot companies.
- Task 5.2: Standardize data formats, API specifications, and KPI dashboards.
- Task 5.3: Validate integration through pilot demonstrations.

Methodologies: API-first design, data schema standardization, API versioning, middleware for data exchange.

Milestones:

Milestone 5.1: API specs defined (Month 18).

Milestone 5.3: KPI dashboards deployed and validated (Month 24).

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	Work package 6: Pilot deployment and validation • Task 6.1: Conduct field tests under real production conditions; collect baseline data. • Task 6.2: Execute end-to-end validation across edge platform, QA/Knowledge, training, and interoperability. • Task 6.3: Compile case studies, ROI estimates, and scalability recommendations. Methodologies: field trials, controlled pilots, statistical evaluation, ROI modeling, cross-site learning. Milestones: Milestone 6.1: End-to-end QA/Knowledge workflow validated in pilots (Month 30). Milestone 6.2: Training effectiveness demonstrated in pilots (Month 36). Milestone 6.3: End-to-end pilots completed (Month 36). Milestone 6.4: Validation report with scaling plan (Month 36).

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Technical contributions: edge-vision platform, QA/Knowledge modules, interoperability interfaces and data driven user needs assessment. Research scope: vision-based quality assistance in manual assembly; edge computing; knowledge extraction and preservance. Resources provided: SOTA AI models, tech stack with ai training hardware, CI/CD, data driven user needs assessment devices. Key tasks: architecture, ai model development, QA/knowledge implementation, process deviation detection, integrations, pilots, compliance, user interaction.
Expected R&R of Korean SME Partner	Process expertise: document current assembly lines, bottlenecks, SOPs, best practice evaluation. Data access: provide available data, access for data recording, MES/ERP feeds, permission to label data. Pilot readiness: 1–3 sites. Change management: assign a change agent, engage users, participate in trainings. Data governance: agree on privacy, ownership, anonymization, audits. Co-development: assist with MES/ERP integrations, APIs, test environments. Resources: allocate staff time, schedules, budget notes. Documentation: co-create SOPs, training materials, case studies. ROI and KPIs: establish baselines, track pilot metrics, provide validation data.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D program will deliver transformative technological achievements that directly address the needs of Korean SMEs in various sectors. At its core, the project will develop an advanced edge-vision platform capable of real-time scene analysis including pose estimation, activity recognition, and object detection. This platform will be complemented by robust Quality Assurance and Knowledge Management modules featuring automated deviation detection and a scalable knowledge base and training system. To ensure broad adoption and seamless integration, the project will define APIs for easy integration, enabling deployment across the piloting SMEs. All components will be underpinned by proven, privacy-preserving data governance models and compliance frameworks, safeguarding data while enabling compliant, scalable innovation. From a commercialization standpoint, the initiative targets a scalable deployment model finely tuned to Korean SMEs, with a clear ROI pathway demonstrated through improvements in First Pass Yield, throughput, and training time reductions. The edge solutions will be market-ready and structured with a modular architecture (Quality Check, Knowledge Base, and Training) that can be adapted to different industries and company sizes. The project will validate customer value through concrete pilot case studies, establishing compelling use-cases that accelerate technology transfer, licensing, or co-development arrangements with industry partners. The broader industrial and economic impacts are substantial. By lifting productivity and competitiveness, the project will enhance the global export readiness and resilience of Korean SMEs. It will drive workforce transformation through on-the-job training, upskilling, and the dissemination of standardized best practices, creating a more skilled manufacturing labor pool. Accelerating digitalization and AI adoption across manufacturing will align with national policy goals and strengthen the innovation ecosystem, generating knowledge spillovers and expanding opportunities for future research funding and public-private collaboration. In sum, the program aims to deliver durable technological capabilities, catalyze widespread SME adoption, and contribute to sustained economic growth and global competitiveness for Korea.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale

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	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization